THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 8: **Playing with Constructions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two circles have radii 8 cm and 5 cm and their centres are 10 cm apart. The circles: 1

(a) touch externally **(b)** do not touch (separate)
(c) intersect at two points **(d)** touch internally

2. A square has a diagonal that splits it into two triangles. If the side is 5 cm, the sum of the angles of one triangle is (degrees): 1

(a) 540 **(b)** 90 **(c)** 180 **(d)** 270

3. A rectangle's diagonals meet at O making $\angle AOB = 50^\circ$. Then $\angle AOD$ is (degrees): 1

(a) 140 **(b)** 160 **(c)** 130 **(d)** 120

4. A circle has diameter 24 cm. A concentric circle has radius 7 cm. The width of the ring (cm) is: 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 4

5. A square has perimeter 32 cm. Its diagonal splits it into two triangles each with two sides of length (cm): 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 24

6. Four equal circles of radius 3 cm are drawn touching in a row. The length of the row (cm) is: 1

(a) 36 **(b)** 24 **(c)** 27 **(d)** 21

7. A rectangle ABCD has AB = 14 cm. M is the midpoint of AB. AM (cm) is: 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 21

8. Two circles have radii 9 cm and 4 cm and their centres are 13 cm apart. The circles: 1

(a) do not touch (separate) **(b)** touch internally
(c) touch externally **(d)** one lies inside the other

9. In rectangle ABCD, AB = 10 cm and BC = 7 cm. AC is a diagonal. The perimeter of rectangle (cm) is: 1

(a) 34 **(b)** 41 **(c)** 44 **(d)** 24

10. A rectangle's diagonals meet at O making $\angle AOB = 70^\circ$. Then $\angle AOD$ is (degrees): 1

(a) 100 **(b)** 120 **(c)** 140 **(d)** 110

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Two circles with radii 9 cm and 4 cm touch externally. Find the distance between their centres, and if they touch internally. 2

12. In rectangle ABCD, $\angle AOB = 40^\circ$ at the point O where the diagonals meet. Find $\angle OAB$ and $\angle AOD$. 2

13. Find the area of a rectangle constructed with sides 10 cm and 7 cm and the area of a square of side 8 cm. Which is larger? 2

14. Two circles with radii 12 cm and 5 cm touch externally. Find the distance between their centres, and if they touch internally. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A square of side 6 cm has its diagonals drawn. Find the area of one of the four triangles formed (it is a quarter of the square) and the sum of angles at the centre. 3

16. Three circles of radius 3 cm touch each other in a row. Find the distance between the centres of the first and third circles, and the total length of the row. 3

17. A rectangle has length 14 cm and its perimeter is 40 cm. Find its breadth and say whether it can be a square. 3

OR

A square of side 8 cm has its diagonals drawn. Find the area of one of the four triangles formed (it is a quarter of the square) and the sum of angles at the centre. 3

18. Three circles of radius 4 cm touch each other in a row. Find the distance between the centres of the first and third circles, and the total length of the row. 3

OR

A rectangle has length 15 cm and its perimeter is 60 cm. Find its breadth and say whether it can be a square. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write the steps to construct a square ABCD of side 4 cm using only ruler and compass. After constructing, state the lengths of both diagonals' halves if the full diagonal is 6 cm and find the angle formed by the diagonals. 5

OR

A gardener draws two concentric circles of radii 5 m and 8 m. Find the width of the path between them, the diameter of the outer circle and how many such paths of the same width fit side by side across the outer diameter (to the nearest whole number). 5

20. Write the steps to construct a square ABCD of side 5 cm using only ruler and compass. After constructing, state the lengths of both diagonals' halves if the full diagonal is 8 cm and find the angle formed by the diagonals. 5

OR

A gardener draws two concentric circles of radii 6 m and 10 m. Find the width of the path between them, the diameter of the outer circle and how many such paths of the same width fit side by side across the outer diameter (to the nearest whole number). 5

END OF QUESTION PAPER — Total: 40 Marks